WORKED SOLUTION

STEP II 2011, Question 1

Pure • Equations with Square Roots

STEP II 2011, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q1

The key idea

Both parts are about square-root sums whose graphs are easy to understand once you find the domain, the endpoint values and the symmetry. In part (i) the picture gives an inequality, the curve never dips below a certain height, and the line can only reach that height at one point, so there is nothing to solve. In part (ii) the graph tells us how many roots to expect, and that lets us throw away the extra root that squaring brings in.

Solution

Part (i)

Let $f(x) = \sqrt{1-x} + \sqrt{3+x}$. Both roots are real only when $1 - x \geq 0$ and $3 + x \geq 0$, so the domain is $-3 \leq x \leq 1$. At the ends, $f(-3) = 2 + 0 = 2$ and $f(1) = 0 + 2 = 2$.

Writing $x = -1 + t$ gives $f = \sqrt{2-t} + \sqrt{2+t}$, which is unchanged when t is replaced by $-t$, so the curve is symmetric about $x = -1$. Differentiating,

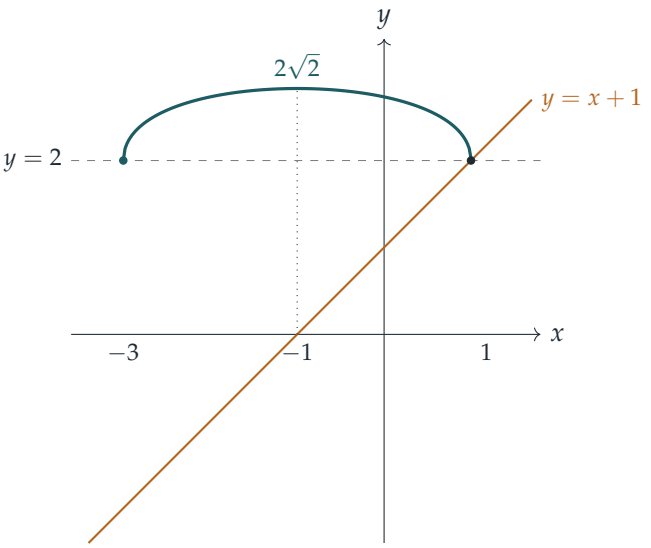
$$f'(x) = -\frac{1}{2\sqrt{1-x}} + \frac{1}{2\sqrt{3+x}},$$

which is zero only when $1 - x = 3 + x$, that is at $x = -1$, where $f(-1) = 2\sqrt{2}$. For $x < -1$ we have $3 + x < 1 - x$, so $f'(x) > 0$; for $x > -1$, $f'(x) < 0$. So f rises from 2 to a maximum of $2\sqrt{2}$ and falls back to 2. Near $x = -3$ the second term of f' blows up, so the curve leaves $(-3, 2)$ vertically, and likewise it arrives at $(1, 2)$ vertically. Each term of f is concave, so the whole curve is a smooth cap.

A cleaner way to see the lowest value is to square:

$$f(x)^2 = (1-x) + (3+x) + 2\sqrt{(1-x)(3+x)} = 4 + 2\sqrt{(1-x)(3+x)} \geq 4,$$

so $f(x) \geq 2$ on the whole domain, with equality only at $x = -3$ and $x = 1$.



Now bring in the line $y = x + 1$. On the domain $x \leq 1$, so $x + 1 \leq 2$, with equality only at $x = 1$. Meanwhile $f(x) \geq 2$. So $f(x) = x + 1$ forces both sides to equal 2, which happens only at $x = 1$. Checking: $\sqrt{0} + \sqrt{4} = 2 = 1 + 1$. So exactly one real value satisfies the equation:

$$x = 1.$$

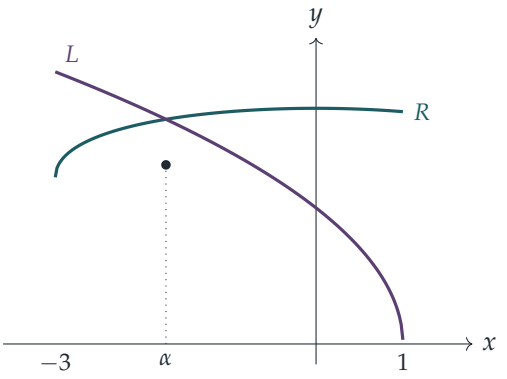
(Outside $[-3, 1]$ the left-hand side isn't defined, so there are no other candidates.)

Part (ii)

Again we need $x \leq 1$, $x \geq -3$ and $x \leq 3$, so the domain is $[-3, 1]$. Compare

$$L(x) = 2\sqrt{1 - x} \qquad \text{and} \qquad R(x) = \sqrt{3 + x} + \sqrt{3 - x}.$$

L is strictly decreasing, from $L(-3) = 4$ to $L(1) = 0$. R has the same shape as the curve in part (i): it is symmetric about $x = 0$ (swap x and $-x$), with $R(\pm 3) = \sqrt{6}$ and a maximum $R(0) = 2\sqrt{3}$, increasing on $[-3, 0]$ and decreasing on $[0, 3]$. On our domain we see only the part from $x = -3$ to $x = 1$, where $R(1) = 2 + \sqrt{2}$.



At $x = -3$, $L = 4 > \sqrt{6} = R$; at $x = 0$, $L = 2 < 2\sqrt{3} = R$. On $[-3, 0]$, L is strictly decreasing and R increasing, so $L - R$ is strictly decreasing and changes sign exactly once. On $[0, 1]$, $L \leq 2$ while $R \geq 2 + \sqrt{2}$, so there is no crossing. Hence there is **exactly one** real solution, and it lies in $-3 < x < 0$.

To find it, square. Both sides are non-negative, so squaring loses nothing:

$$\begin{aligned} & 4(1-x) = (3+x) + (3-x) + 2\sqrt{9-x^2} \\ \iff \quad & -1 - 2x = \sqrt{9-x^2}. \end{aligned}$$

Ask yourself When we square again, which solutions of the new equation are genuine?

The right-hand side is a square root, so it is ≥ 0; a genuine solution needs $-1 - 2x \geq 0$, that is $x \leq -\frac{1}{2}$. With that condition, squaring is reversible:

$$1 + 4x + 4x^2 = 9 - x^2 \iff 5x^2 + 4x - 8 = 0 \iff x = \frac{-2 \pm 2\sqrt{11}}{5}.$$

The root $\frac{-2+2\sqrt{11}}{5} \approx 0.93$ breaks $x \leq -\frac{1}{2}$, so we reject it. The other root, $\frac{-2-2\sqrt{11}}{5} \approx -1.73$, satisfies it and lies in $[-3, 0]$, as the graph predicted. So the unique solution is

$$x = -\frac{2 + 2\sqrt{11}}{5}.$$

Checks

- At $x \approx -1.72665$: $2\sqrt{1-x} \approx 3.3025$ and $\sqrt{3+x} + \sqrt{3-x} \approx 1.1284 + 2.1741 \approx 3.3025$. The two sides agree.
- The rejected root $x \approx 0.92665$ gives $L \approx 0.542$ but $R \approx 3.42$. It actually solves $\sqrt{3+x} - \sqrt{3-x} = 2\sqrt{1-x}$, which squares to the same thing.
- In part (i), $f(-1) = 2\sqrt{2} \approx 2.83 > 0 = (-1) + 1$, consistent with the line lying below the curve everywhere except at $(1, 2)$.

Common slips

- In part (i), squaring and solving algebraically, which is messy and can bring in spurious roots, when the bound $f \geq 2 \geq x + 1$ settles it at once.
- Keeping both roots of $5x^2 + 4x - 8 = 0$. Every squaring step needs its sign condition, here $-1 - 2x \geq 0$.
- Forgetting the domain: neither equation means anything for $x > 1$.

Beyond STEP

Part (i) is a small example of a common analysis technique: rather than solving $f(x) = g(x)$, show that $f \geq c \geq g$ and look at when equality holds. Part (ii) uses the intermediate value theorem together with monotonicity to count roots before finding any, which is how existence and uniqueness are proved in a first analysis course. Abbott's *Understanding Analysis* gives a careful treatment of continuity and the intermediate value theorem.

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